

Customer Churn Analysis and Prediction

Internship Project Report

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Internship Domain: Data Science

Submission Date: 06/07/2026

Executive Summary

Customer churn is a major concern for telecommunication companies because losing customers leads to reduced revenue and increased customer acquisition costs. This project analyzed customer behavior using the Telco Customer Churn dataset and developed machine learning models to predict customers who are likely to discontinue the company's services.

The project involved data preparation, exploratory data analysis (EDA), customer segmentation, predictive modeling, model evaluation, and business recommendations. Logistic Regression achieved the best performance after hyperparameter tuning, with an accuracy of 80.55% and an AUC score of 84.11%. The findings showed that contract type, tenure, monthly charges, payment method, and internet service significantly influence customer churn. Based on these insights, practical recommendations were proposed to improve customer retention and reduce churn.

1. Project Overview

Objective

The aim of this project was to analyze customer churn patterns, identify the factors influencing customer attrition, and develop a predictive model capable of identifying customers at risk of churning.

Dataset

The Telco Customer Churn dataset contains customer demographic information, subscribed services, account details, billing information, and churn status. After data cleaning, categorical variables were encoded for machine learning, and the dataset was split into training and testing sets.

Tools Used

- Python

- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

2. Methodology

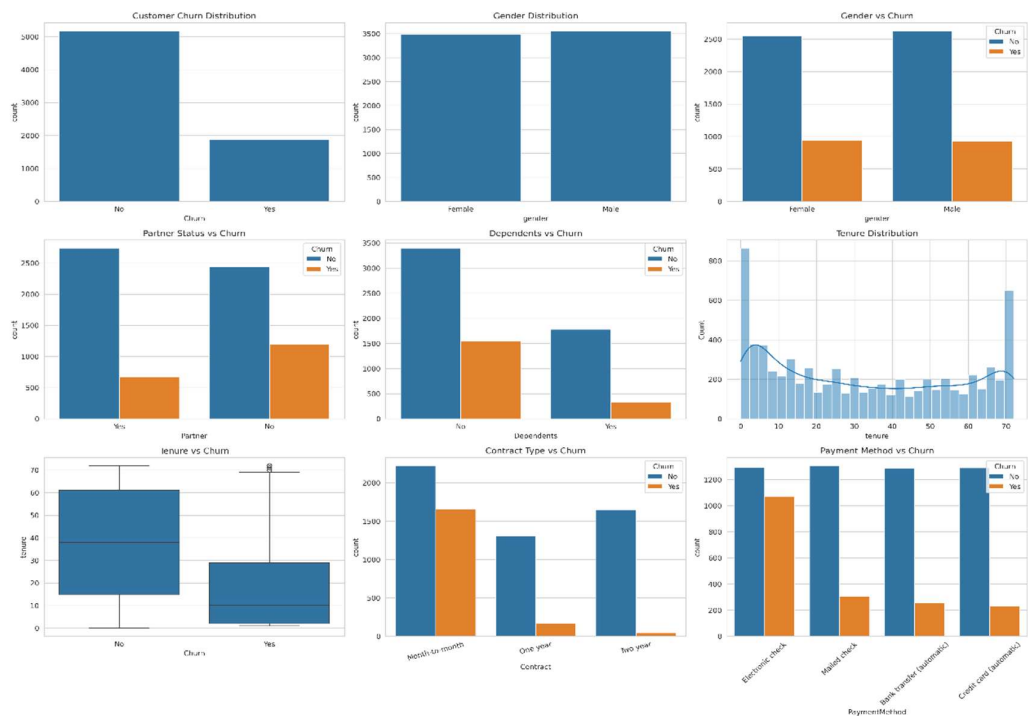
The project followed a structured data science workflow:

Task 1 – Data Preparation

- Loaded and explored the dataset.
- Handled missing values in the TotalCharges column.
- Encoded categorical variables using one-hot encoding.
- Split the dataset into training and testing sets.

Task 2 – Exploratory Data Analysis

EDA was conducted to examine churn patterns across customer demographics, tenure, contract type, monthly charges, and payment methods.

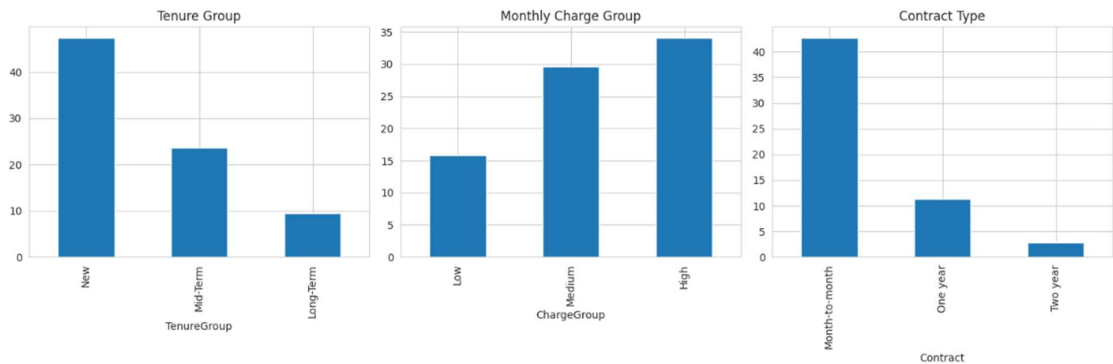


Task 3 – Customer Segmentation

Customers were segmented based on:

- Tenure
- Monthly Charges
- Contract Type

High-value customers at risk of churn were also identified.



3. Results and Discussion

Exploratory Data Analysis

The analysis revealed several important patterns:

- Customers on month-to-month contracts exhibited the highest churn rate.
- New customers were more likely to churn than long-term customers.
- Customers with higher monthly charges experienced greater churn.
- Payment methods and contract types significantly influenced customer retention.

Customer Segmentation

Key segmentation findings include:

- **New customers:** 47.44% churn rate
- **High monthly charge customers:** 34.09% churn rate
- **Month-to-month contracts:** 42.71% churn rate
- **High-value customer churn:** 15.16%

These findings indicate that retention strategies should prioritize customers in these segments.

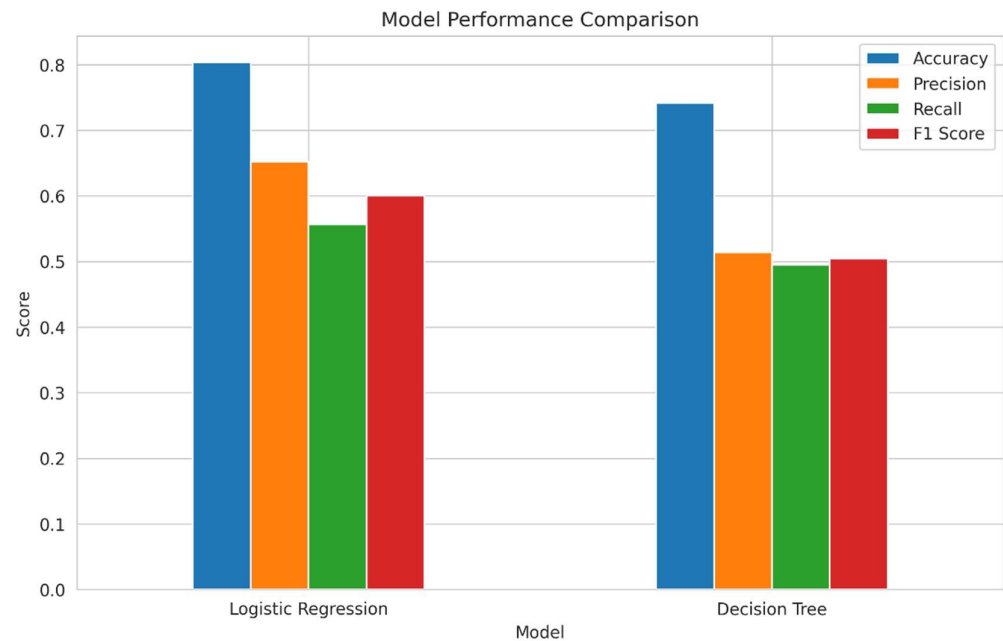
Predictive Modeling

Two machine learning models were developed:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80.34%	65.20%	55.61%	60.03%
Decision Tree	74.17%	51.39%	49.47%	50.41%

After hyperparameter tuning using GridSearchCV, the optimized Logistic Regression model achieved:

- **Accuracy:** 80.55%
- **Precision:** 65.72%
- **Recall:** 55.88%
- **F1 Score:** 60.40%



4. Model Evaluation

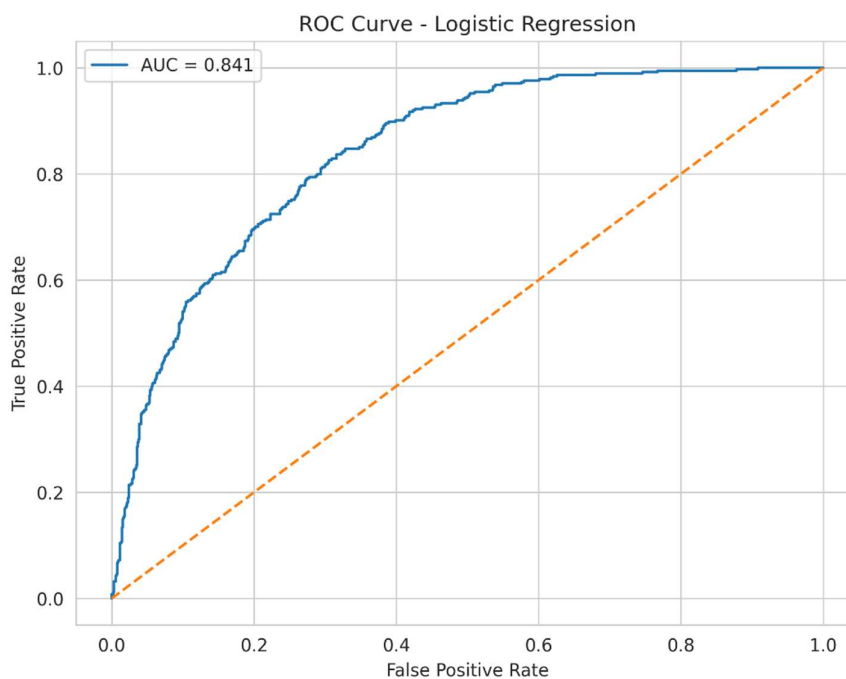
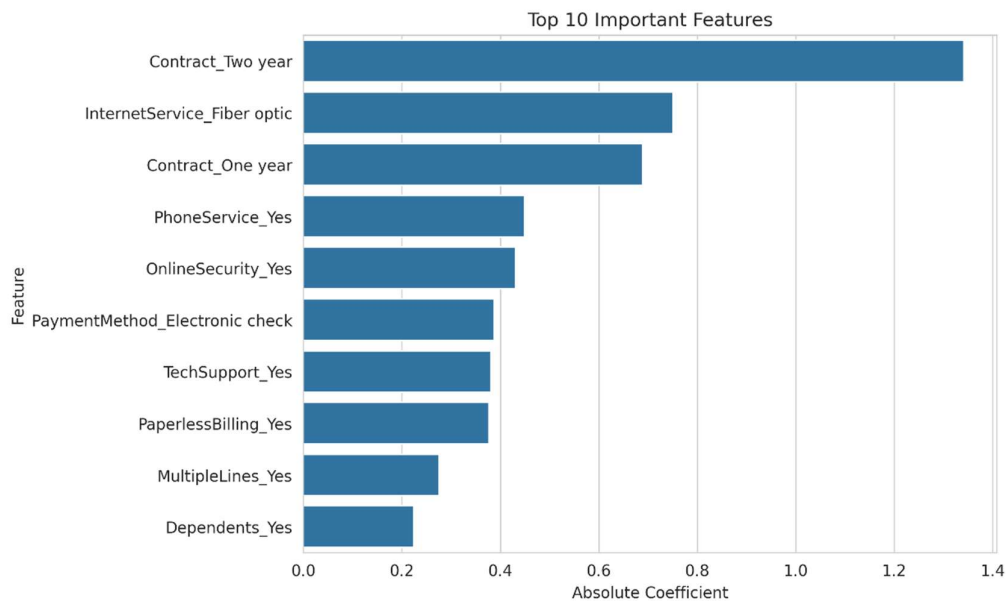
Feature importance analysis showed that the strongest predictors of customer churn were:

- Contract type
- Internet service
- Payment method
- Online security

- Technical support

Customers on one-year and two-year contracts were significantly less likely to churn, while customers using fiber optic internet and electronic check payment methods were more likely to leave.

The optimized Logistic Regression model achieved an AUC score of 84.11%, indicating strong predictive performance.



5. Business Recommendations

Based on the findings, the following recommendations are proposed:

- Encourage month-to-month customers to migrate to long-term contracts through loyalty incentives.
- Strengthen onboarding programs to improve retention among new customers.
- Review pricing strategies and provide personalized offers for high-paying customers.
- Promote value-added services such as Online Security and Technical Support.
- Deploy the predictive model to proactively identify and engage customers at risk of churning.

These recommendations are expected to reduce churn, improve customer satisfaction, and increase long-term revenue.

6. Conclusion

This project successfully analyzed customer churn and developed a predictive model capable of identifying customers at risk of leaving the company. The optimized Logistic Regression model achieved 80.55% accuracy and an AUC score of 84.11%, demonstrating strong predictive capability.

The analysis identified contract type, customer tenure, monthly charges, payment method, internet service, online security, and technical support as the most influential factors affecting churn. By implementing the recommended retention strategies and leveraging predictive analytics, the company can improve customer loyalty, reduce churn, and support sustainable business growth.